Adventures in Covariance

EES 4891-06/5891-01

Bayesian Statistical Methods

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Adventures in Covariance

Varying Intercepts and Slopes

- Coffee Robot Example
  - A robot visits N cafes
  - Estimate the average waiting time W to get a coffee.
  - Model:

$$
\begin{aligned}
W_{\text{cafe}} &\sim \text{Normal}(\mu_{\text{cafe}}, \sigma) \\
\mu_{\text{cafe}} &= \alpha_{\text{cafe}} \\
\alpha_{\text{cafe}} &\sim \text{Normal}(5, 2) \\
\sigma &\sim \text{Exponential}(1)
\end{aligned}
$$

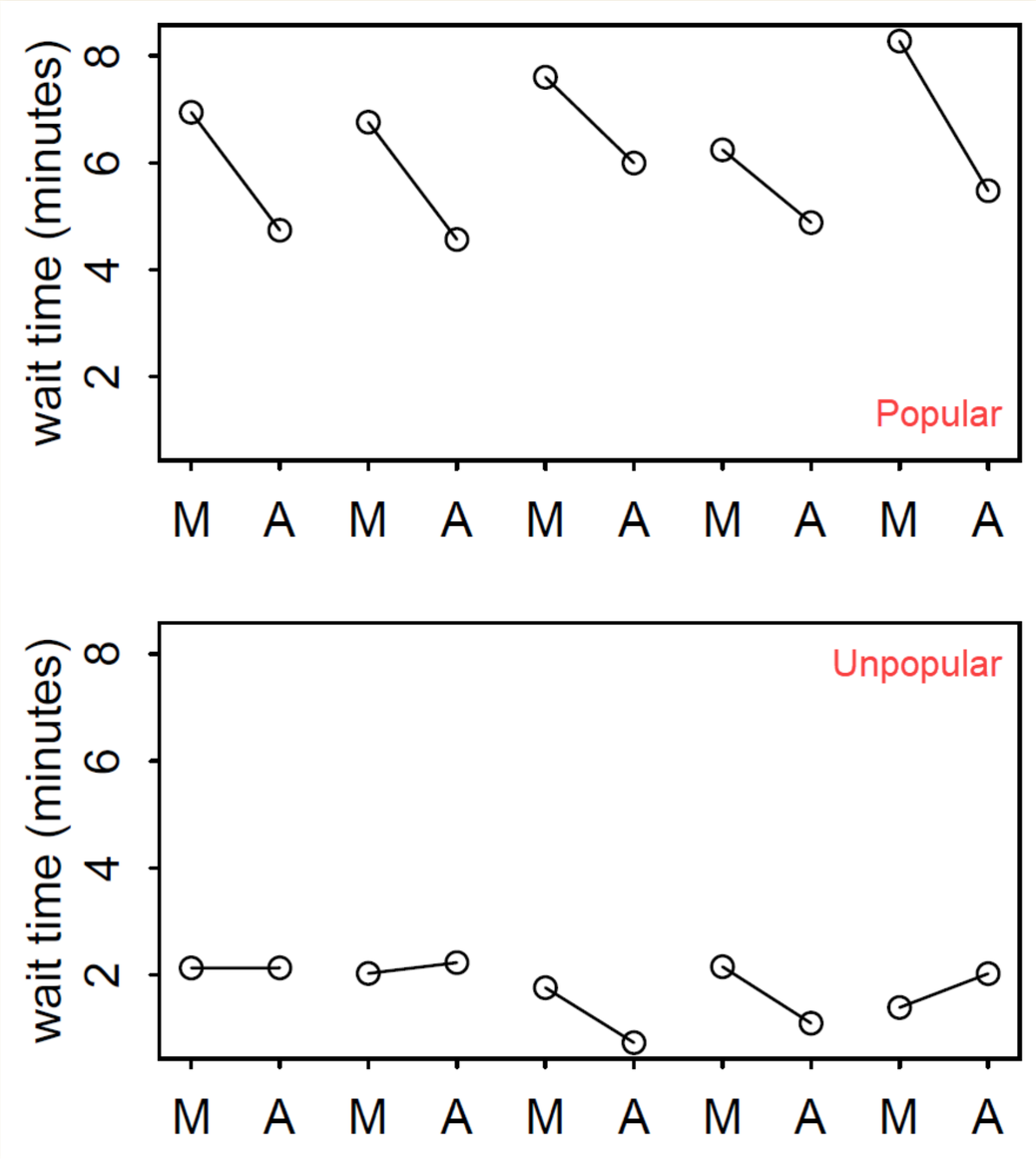
- Different times of day
  - Morning (busier) vs. afternoon (less busy)
  - Model:

$$
\begin{aligned}
W_{\text{cafe}} &\sim \text{Normal}(\mu_{\text{cafe}}, \sigma) \\
\mu_{\text{cafe}} &= \alpha_{\text{cafe}} + \beta_{\text{cafe}} A \\
\alpha_{\text{cafe}} &\sim \text{Normal}(5, 2) \\
\beta_{\text{cafe}} &\sim \text{Normal}(-1, 0.5) \\
\sigma &\sim \text{Exponential}(1)
\end{aligned}
$$

  - Something's missing ...

Correlations

- Some cafes are very popular, and others are not popular
- Popular cafes are busy
  - Big differences between morning and afternoon
- Unpopular cafes aren't busy
  - Not much difference between morning and afternoon
- **Covariance** between *slopes* (β) and *intercepts* (α).



Modeling covariance

- Covariance matrix

$$\begin{pmatrix} \text{variance of intercepts} & \text{covariance of intercepts and slopes} \\ \text{covariance of intercepts and slopes} & \text{variance of slopes} \end{pmatrix}$$

$$\begin{pmatrix} \sigma_\alpha^2 & \sigma_\alpha \sigma_\beta \rho_{\alpha\beta} \\ \sigma_\alpha \sigma_\beta \rho_{\alpha\beta} & \sigma_\beta^2 \end{pmatrix}$$

  - $\rho_{\alpha\beta}$ is the correlation between slopes and intercepts ($-1 \leq \rho_{\alpha\beta} \leq 1$).

Simulation modeling

Simulation modeling

- When we fit our model to observed data, we don't know the true parameter values.
  - We can't assess how well the model estimates are.
- Before fitting observed data, generate *simulated data* using known parameters and see how accurate the models' estimates are.
- Simulation parameters

```
a <- 3.5              # average morning wait time
b <- -1               # average difference afternoon wait time
sigma_a <- 1          # std dev in intercepts
sigma_b <- 0.5        # std dev in slopes
rho <- -0.7           # correlation between intercepts and slopes

Mu <- c( a, b)
```

- Covariance

```
cov_ab <- sigma_a * sigma_b + rho
Sigma <- matrix(c(sigma_a^2, cov_ab, cov_ab, sigma_b^2), ncol =
        2)
```

- Generate α and β params with covariance

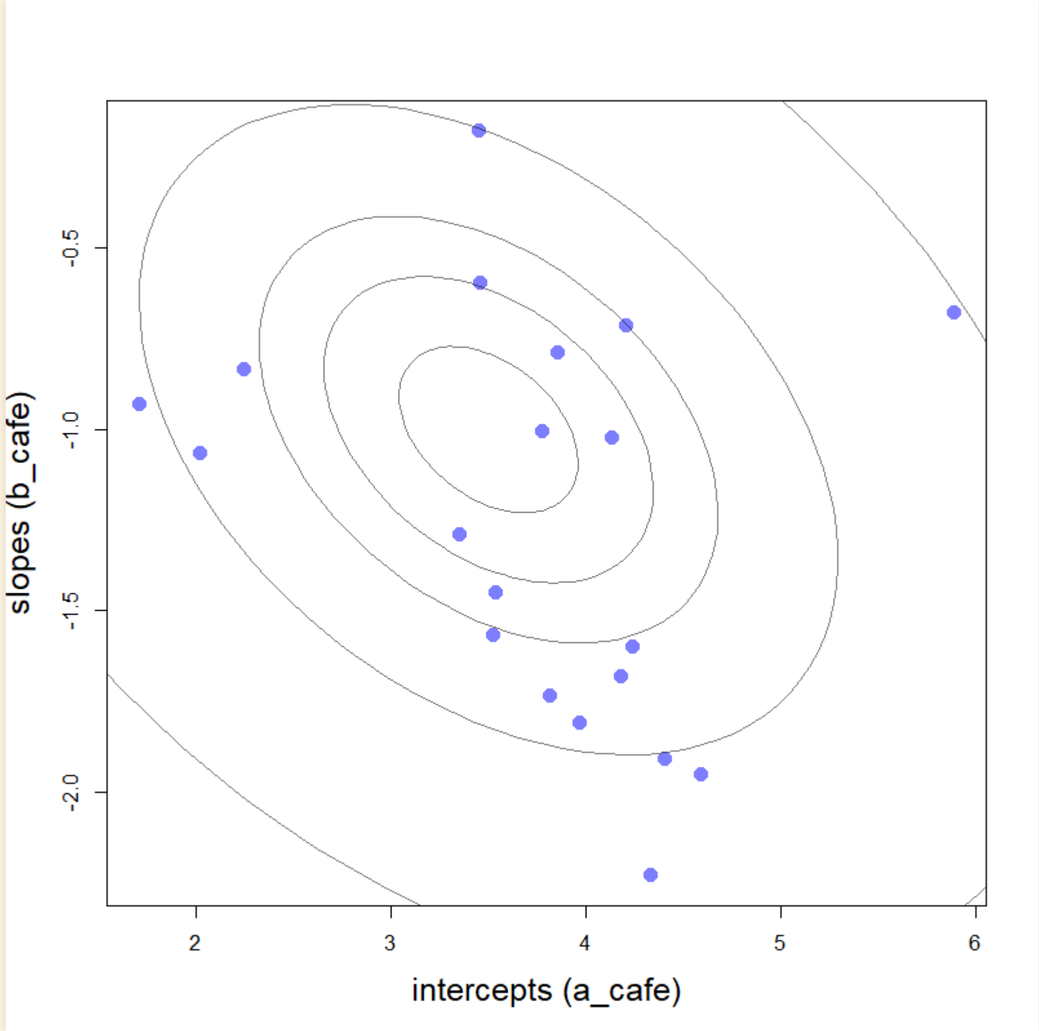
```
library(MASS)

set.seed(5) # used to replicate example
N_cafes <- 20
vary_effects <- mvrnorm(N_cafes, Mu, Sigma)

a_cafe <- vary_effects[,1]
b_cafe <- vary_effects[,2]
```

- Generate simulated data

```
set.seed(22)
N_visits <- 10
afternoon <- rep(0:1, N_visits * N_cafes / 2)
cafe_id <- rep(1:N_cafes , each = N_visits)
mu <- a_cafe[cafe_id] + b_cafe[cafe_id] * afternoon
sigma <- 0.5  # std dev within cafes
wait <- rnorm(N_visits * N_cafes, mu, sigma)
d <- data.frame(cafe = cafe_id, afternoon = afternoon, wait =
        wait)
```

What Is Covariance?

```
library(ellipse)

plot(a_cafe, b_cafe, col = rangi2, pch=19, cex=1.5, cex.lab=1.5,
    xlab="intercepts (a_cafe)", ylab="slopes (b_cafe)")

# overlay population distribution
for ( l in c(0.1,0.3,0.5,0.8,0.99) )
    lines(ellipse(Sigma, centre=Mu, level=l),
          col = col.alpha("black",0.5), lwd=1)
```



Fitting the Model:
Varying Slopes & Intercepts

Fitting the Model

Model

$$W \sim \text{Normal}(\mu, \sigma)$$
$$\mu = \alpha_{\text{cafe}} + \beta_{\text{cafe}} A$$
$$\begin{bmatrix} \alpha_{\text{cafe}} \\ \beta_{\text{cafe}} \end{bmatrix} \sim \text{MVNormal}\left(\begin{bmatrix} \alpha \\ \beta \end{bmatrix}, S \right)$$
$$S = \begin{pmatrix} \sigma_\alpha & 0 \\ 0 & \sigma_\beta \end{pmatrix} R \begin{pmatrix} \sigma_\alpha & 0 \\ 0 & \sigma_\beta \end{pmatrix}$$

Priors

$$\alpha \sim \text{Normal}(5, 2)$$
$$\beta \sim \text{Normal}(-1, 0.5)$$
$$\sigma \sim \text{Exponential}(1)$$
$$\sigma_\alpha \sim \text{Exponential}(1)$$
$$\sigma_\beta \sim \text{Exponential}(1)$$
$$R \sim \text{LKJcorr}(2)$$

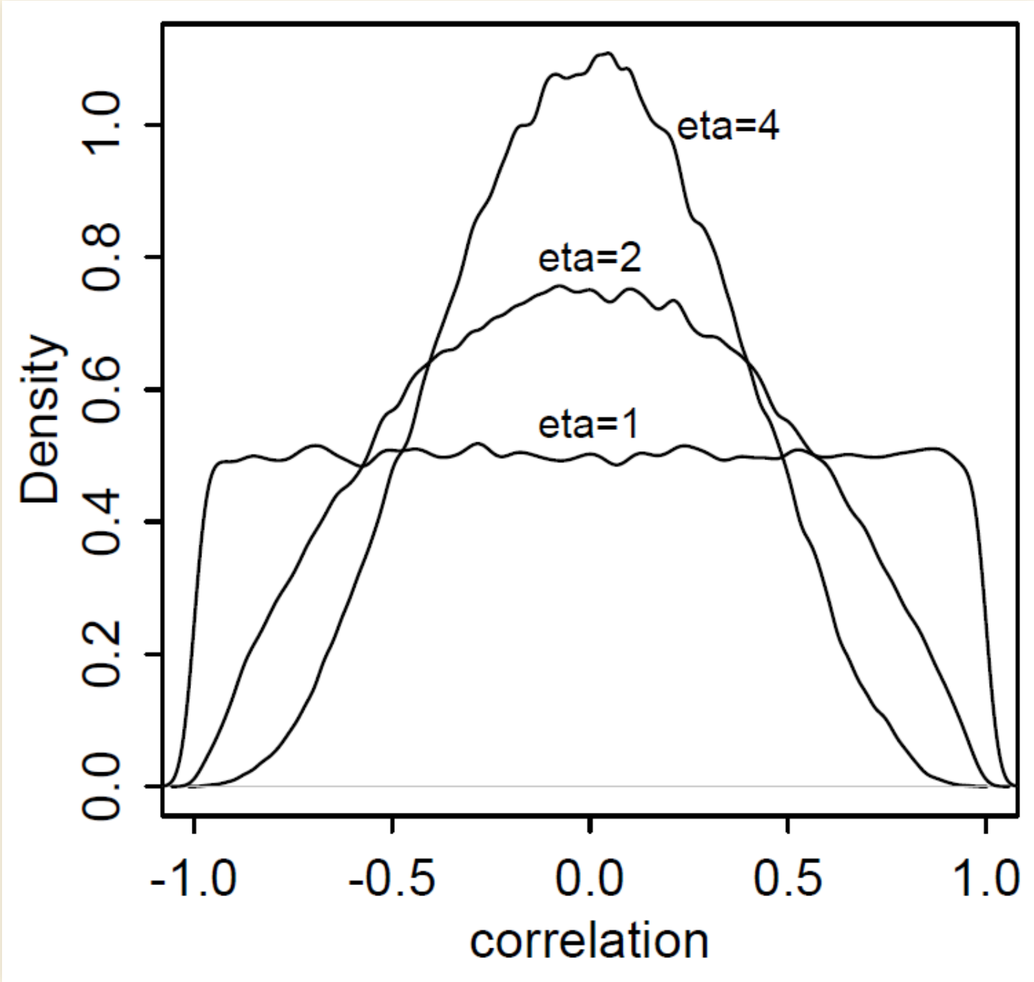
- LKJ prior:
  - Weakly-informative prior for correlation matrices.

LKJ Priors

- Simple correlation matrix:

$$R = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$$

- LKJcorr(η)
  - A probability distribution over different values of ρ
  - $\eta \geq 1$
    - The greater η is, the more unlikely extreme correlations (ρ close to ± 1) are
  - $\eta = 2$ is a common default prior.



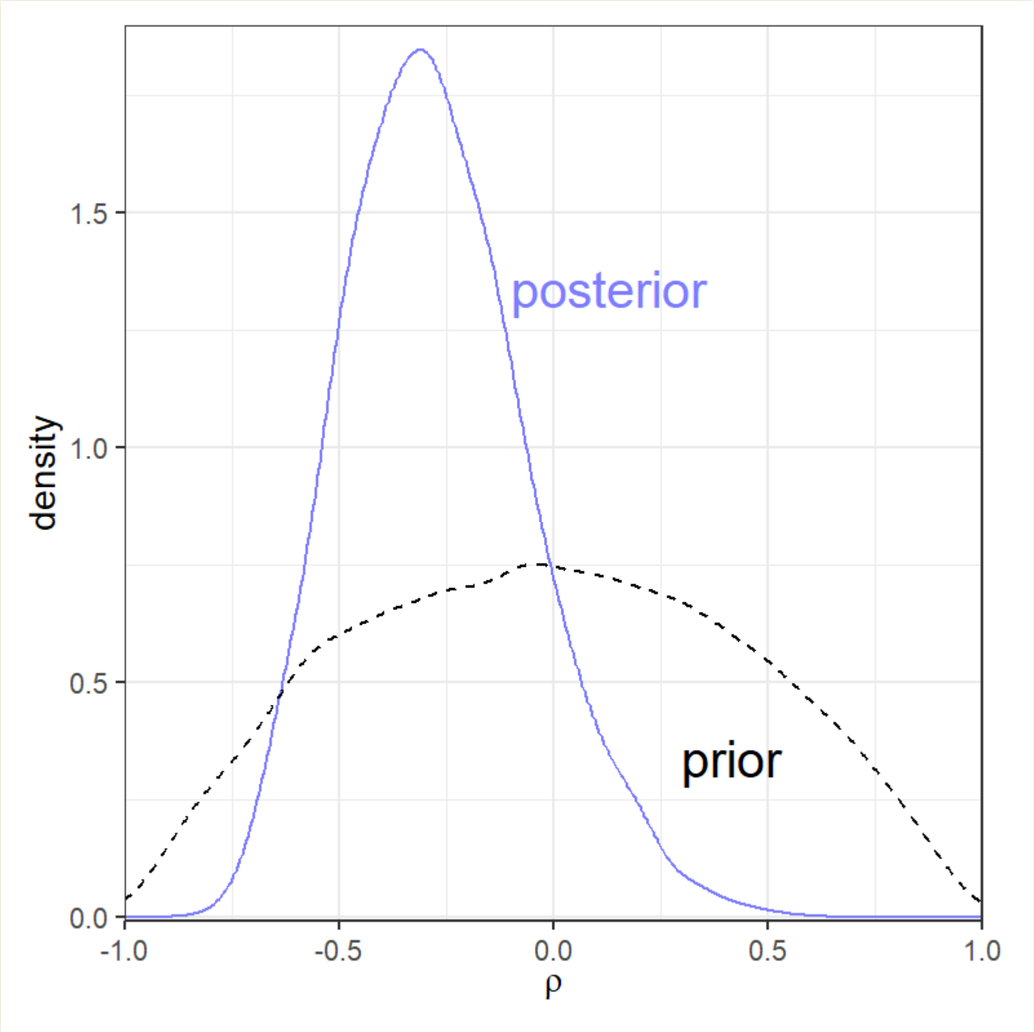
Fitting the model

- Model

```
set.seed(867530)
mdl_cafe <- ulam(
    alist(
        wait ~ normal(mu, sigma),
        mu <- a_cafe[cafe] + b_cafe[cafe] * afternoon,
        c(a_cafe,b_cafe)[cafe] ~ multi_normal(c(a, b), Rho,
         sigma_cafe),
        a ~ normal(5, 2),
        b ~ normal(-1, 0.5),
        sigma_cafe ~ exponential(1),
        sigma ~ exponential(1),
        Rho ~ lkj_corr(2)
    ), data=d, chains = 4, cores = 4)
```

- Extract samples from posterior

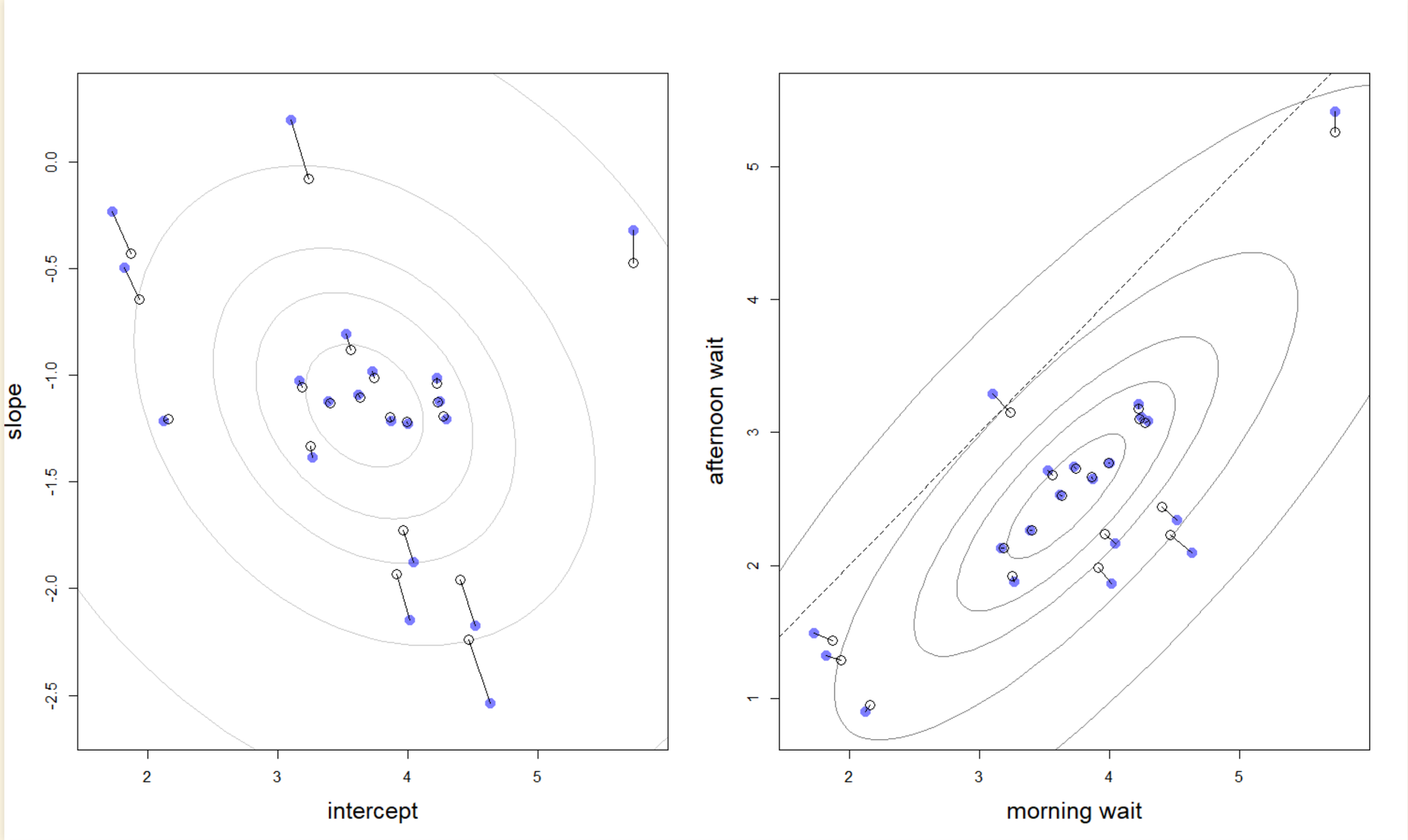
```
post <- tidy_draws(mdl_cafe) %>% mutate(label = "posterior") %>%
  rename(rho = `Rho[1,2]`)
prior <- tibble(rho = rlkjcorr(1e4, K=2, eta=2)[,1,2], label = "prior")
labels <- tibble(rho = c(-0.1, 0.3), y = c(1.3, 0.3),
                 label = c("posterior", "prior"))

ggplot(post, aes(x = rho, color = label)) +
geom_density(bw = 0.05, linewidth=1) +
geom_density(data = prior, linetype="dashed") +
geom_text(data = labels, aes(y = y, label = label), size = 10,
          vjust = 0, hjust = 0) +
scale_x_continuous(limits = c(-1,1), expand = c(0,0)) +
scale_y_continuous(expand = expansion(c(0.005, 0), c(0, 0.05))) +
scale_color_manual(values = c(prior = "black", posterior = rangi2),
                   guide = "none") +
labs(x = expr(rho), y = "density") +
theme(plot.margin = unit(c(1,2,1,1), "lines"))
```



- Posterior is well within the prior:
  - The data are determining the posterior

Shrinkage and Regularization

- **Shrinkage**: Partial pooling moves estimates for each cafe toward the average



Advanced Applications

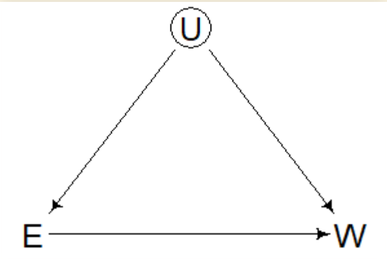
Advanced Applications

- The following examples give you a taste of some ways we apply covariance among the parameters in our models
- The examples should give you a sense of how powerful these tools can be
  - Instrumental Variables
  - Gaussian Processes
- The methods I'm going to show you are very complicated and I will only present a superficial perspective.
  - If you think these will be useful to you, there are more comprehensive accounts you can read to learn more
- Gaussian Processes are very computationally demanding and it can be hard to get them to work well
  - But they can be very powerful and there are tools to make it easier to work with them.

Instrumental Variables

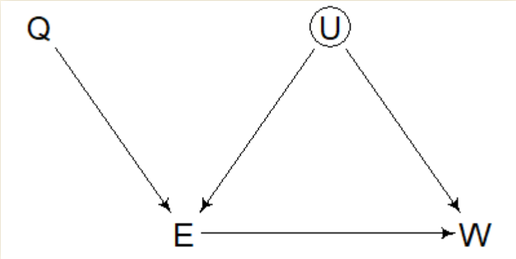
Instrumental Variables

- Effect of education on wages
  - Does more school improve future wages?
  - Complication: People who are motivated and disciplined are likely to stay in school longer, and also earn more regardless of schooling:



  - U is an unobserved variable that describes the person's work ethic and self-discipline.
  - The backdoor path $E \leftarrow U \rightarrow W$ prevents us from drawing causal inferences from correlations between E and W.

- **Instrumental variables:**

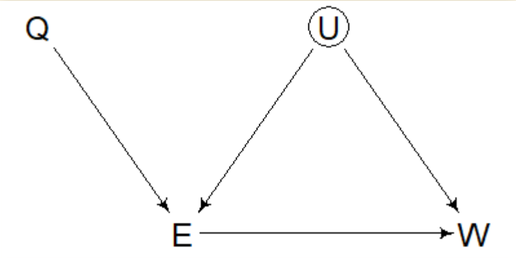


  - Q can be an **instrumental variable** if:
    1. It's independent of U.
    2. It's not independent of E.
    3. Q cannot influence W except through E.

Models with Instrumental Variables

- What are instrumental variables?
- Natural experiments
- We don't have enough control of the system to do *controlled experiments*
- SO we use natural variations in *instrumental variables* as *natural experiments*

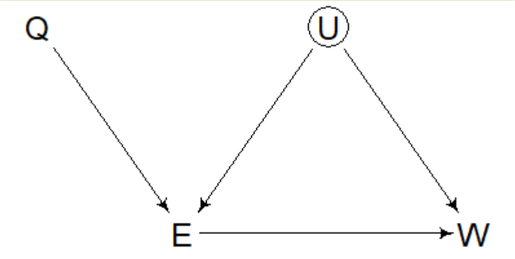
- DAG:



- We can't just put Q in a regression model
  - E is a *collider* of Q and U.
  - The collider opens up a *non-causal* path from Q to W, which creates a false correlation between Q and W, even if E doesn't influence W.
    - Q is a **bias amplifier**.

Example of Instrumental Variable

- Example:



- Q is the season of the year
  - Students born earlier complete fewer grades by the time they reach the age where they can leave school (drop out).
  - Q is not influenced by any of the other variables

- Generate simulated data:

```
set.seed(73)
N <- 500

U_sim <- rnorm(N)
Q_sim <- sample(1:4, size = N, replace = TRUE)
E_sim <- rnorm(N, U_sim + Q_sim)
W_sim <- rnorm(N, U_sim + 0 * E_sim)

dat_sim <- list(
  W=standardize(W_sim),
  E=standardize(E_sim),
  Q=standardize(Q_sim)
)
```

  - Notice that in the simulated data, E has no effect on W.

Writing a model

```
mdl_wages <- ulam(
  alist(
    W ~ dnorm(mu, sigma),
    mu <- aW + bEW * E,
    aW ~ dnorm(0, 0.2),
    bEW ~ dnorm(0, 0.5),
    sigma ~ dexp(1)
  ), data = dat_sim, chains = 4, cores = 4)
```

```
precis_show(precis(mdl_wages, digits = 2))
```

```
##       mean   sd  5.5% 94.5% rhat ess_bulk
## aW    0.00 0.04 -0.06  0.06 1.00  2038.29
## bEW   0.40 0.04  0.33  0.46 1.00  2161.75
## sigma 0.92 0.03  0.88  0.96 1.01  1975.45
```

- The influence of U causes the model to incorrectly think E has a significant effect on W.

```
mdl_wages_q <- ulam(
  alist(
    W ~ dnorm(mu, sigma),
    mu <- aW + bEW * E + bQW * Q,
    aW ~ dnorm(0, 0.2),
    bEW ~ dnorm(0, 0.5),
    bQW ~ dnorm(0, 0.5),
    sigma ~ dexp(1)
  ), data = dat_sim, chains = 4, cores = 4)
```

```
precis_show(precis(mdl_wages_q, digits = 2))
```

```
##        mean   sd  5.5% 94.5% rhat ess_bulk
## aW     0.00 0.04 -0.06  0.06    1  1970.73
## bEW    0.64 0.05  0.56  0.71    1  1247.72
## bQW   -0.40 0.05 -0.48 -0.33    1  1232.88
## sigma  0.86 0.03  0.82  0.90    1  1714.75
```

- This is even worse:
  - The estimate of the effect of E is even larger
  - The model also thinks Q has a direct effect on W.

A Better Model of Instrumental Variables

- *Generative* model:
  - Formula

$$
\begin{aligned}
W &\sim \text{Normal}(\mu_W, \sigma_W) \\
\mu_W &= \alpha_W + \beta_{EW} E + U \\
E &\sim \text{Normal}(\mu_E, \sigma_E) \\
\mu_E &= \alpha_E + \beta_{QE} Q + U \\
Q &\sim \text{Categorical}([0.25, 0.25, 0.25, 0.25]) \\
U &\sim \text{Normal}(0, 1)
\end{aligned}
$$

- *Statistical model*:
  - We can't measure U, so our *statistical model* averages over the different unobserved values.
  - Formula:

$$
\begin{aligned}
\begin{pmatrix} W \\ E \end{pmatrix} &\sim \text{MVNormal}\left(\begin{pmatrix} \mu_W \\ \mu_E \end{pmatrix}, S\right) \\
\mu_W &= \alpha_W + \beta_{EW} E \\
\mu_E &= \alpha_E + \beta_{QE} Q
\end{aligned}
$$

Coding the Better Model

- Formula for *statistical* model:

$$\begin{pmatrix} W \\ E \end{pmatrix} \sim \text{MVNormal}\left(\begin{pmatrix} \mu_W \\ \mu_E \end{pmatrix}, S\right)$$
$$\mu_W = \alpha_W + \beta_{EW} E$$
$$\mu_E = \alpha_E + \beta_{QE} Q$$

- We can't measure U, so our *statistical model* averages over the different unobserved values.

- Model code

```
mdl_wages_inst <- ulam(
  alist(
    c(W,E) ~ multi_normal(c(muW, muE), Rho, Sigma),
    muW <- aW + bEW * E,
    muE <- aE + bQE * Q,
    c(aW,aE) ~ normal(0, 0.2),
    c(bEW,bQE) ~ normal(0, 0.5),
    Rho ~ lkj_corr(2),
    Sigma ~ exponential(1)
  ), data = dat_sim, chains = 4, cores = 4)
```

```
precis_show(precis(mdl_wages_inst, digits = 2, depth=3))
```

```
##            mean   sd  5.5% 94.5% rhat ess_bulk
## aE         0.00 0.04 -0.06  0.06 1.00  1214.03
## aW         0.00 0.05 -0.07  0.07 1.00  1282.72
## bQE        0.59 0.04  0.53  0.65 1.00  1414.31
## bEW       -0.05 0.08 -0.17  0.07 1.01   779.82
## Rho[1,1]   1.00 0.00  1.00  1.00   NA       NA
## Rho[2,1]   0.54 0.05  0.45  0.62 1.01   811.03
## Rho[1,2]   0.54 0.05  0.45  0.62 1.01   811.03
## Rho[2,2]   1.00 0.00  1.00  1.00   NA       NA
## Sigma[1]   1.02 0.05  0.95  1.10 1.01   881.51
## Sigma[2]   0.81 0.03  0.77  0.85 1.00  1349.61
```

- μ_{EW} is consistent with zero, as it ought to be.

New Data

- Generate new data, where E has a positive effect on wages.

```
set.seed(73)
N <- 500

U_sim <- rnorm(N)
Q_sim <- sample(1:4, size = N, replace = TRUE)
E_sim <- rnorm(N, U_sim + Q_sim)
W_sim <- rnorm(N, U_sim + 0.3 * E_sim)

dat_sim_2 <- list(
  W=standardize(W_sim),
  E=standardize(E_sim),
  Q=standardize(Q_sim)
)
```

- Rerun the model with new data:

```
mdl_wages_inst_2 <- ulam(mdl_wages_inst, data = dat_sim_2,
                         chains = 4, cores = 4)
```

```
precis_show(precis(mdl_wages_inst_2, digits = 2, depth=3))
```

```
##          mean   sd  5.5% 94.5% rhat ess_bulk
## aE       0.00 0.04 -0.06  0.06    1  1568.54
## aW       0.00 0.04 -0.06  0.06    1  1611.81
## bQE      0.59 0.04  0.53  0.65    1  1328.21
## bEW      0.23 0.06  0.13  0.33    1  1009.65
## Rho[1,1] 1.00 0.00  1.00  1.00   NA       NA
## Rho[2,1] 0.54 0.05  0.46  0.62    1   999.93
## Rho[1,2] 0.54 0.05  0.46  0.62    1   999.93
## Rho[2,2] 1.00 0.00  1.00  1.00   NA       NA
## Sigma[1] 0.88 0.04  0.82  0.94    1  1052.93
## Sigma[2] 0.81 0.03  0.77  0.85    1  1763.76
```

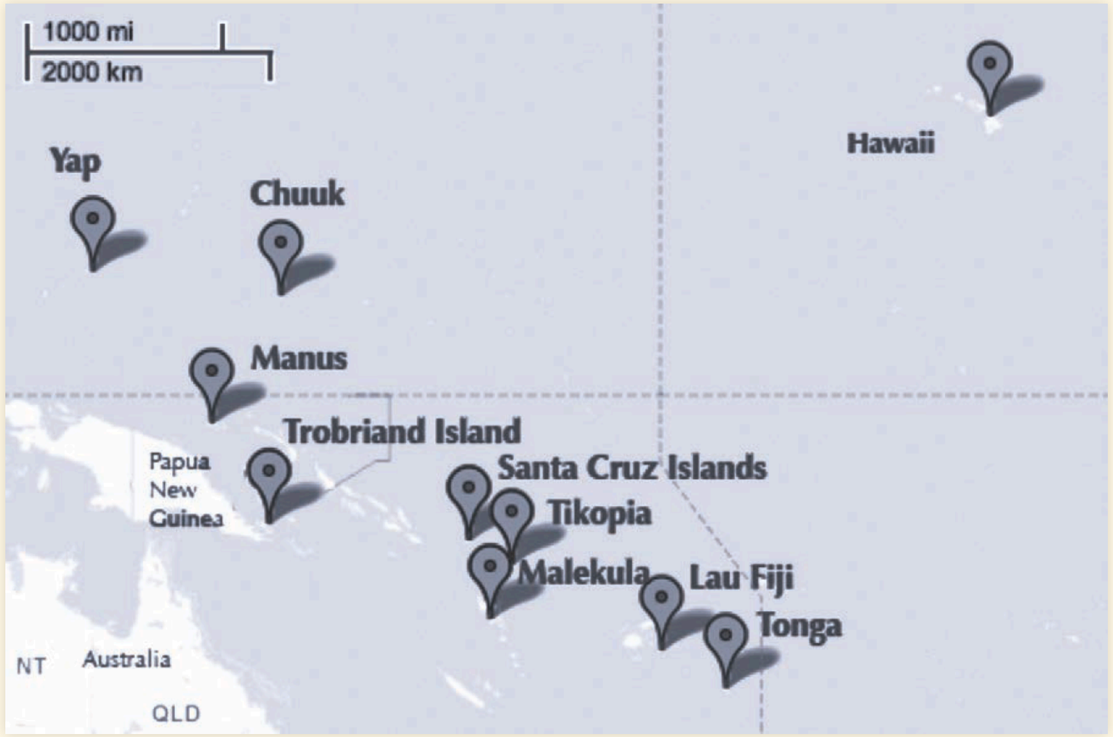
  - β_{EW} is consistent with the correct value 0.3.

Gaussian Processes

Continuous Categories

- Everything we've looked at with multilevel models so far has studied the varying effects of discrete categories (cafes, individual chimpanzees, academic departments, etc.).
  - Data is organized in discrete clusters.
  - There's no special order to the clusters, no significance of being cluster #1.
  - What about clustering on continuously distributed variables?
    - Height
    - Age
    - Income
    - ...
  - **Gaussian Process** models allow us to study variation in slope or intercept for continuously distributed predictor variables.

- Distribution of Tools across the islands of Oceania



The Data

```
library(rethinking)
data(Kline)
data("islandsDistMatrix")

d <- Kline
dist <- islandsDistMatrix

kable(d)
```

culture	population	contact	total_tools	mean_TU
Malekula	1100	low	13	3.2
Tikopia	1500	low	22	4.7
Santa Cruz	3600	low	24	4.0
Yap	4791	high	43	5.0
Lau Fiji	7400	high	33	5.0
Trobriand	8000	high	19	4.0
Chuuk	9200	high	40	3.8
Manus	13000	low	28	6.6
Tonga	17500	high	55	5.4
Hawaii	275000	low	71	6.6

- **culture:** Name of island culture
- **population:** Historical population size
- **contact:** low or high contact rate with other islands
- **total tools**: number of tools in historical tool kit
- **mean TU:** a measure of tool complexity
- Distance matrix (thousands of km):

	Malekula	Tikopia	Santa Cruz	Yap	Lau Fiji	Trobriand
Malekula	0.000	0.475	0.631	4.363	1.234	2.036
Tikopia	0.475	0.000	0.315	4.173	1.236	2.007
Santa Cruz	0.631	0.315	0.000	3.859	1.550	1.708
Yap	4.363	4.173	3.859	0.000	5.391	2.462
Lau Fiji	1.234	1.236	1.550	5.391	0.000	3.219
Trobriand	2.036	2.007	1.708	2.462	3.219	0.000

Modeling Tool Acquisition

- Single-level Model:

$$T \sim \text{Poisson}(\lambda)$$
$$\lambda = \alpha P^{\beta} / \gamma$$

- Multilevel model:

$$T \sim \text{Poisson}(\lambda)$$
$$\lambda = \exp(k_{\text{society}}) \alpha P^{\beta} / \gamma$$

  - k is the varying intercept.
  - Instead of assigning it randomly to each society category, we're going to try to predict it from the distance of other islands.

$$\begin{pmatrix} k_1 \\ k_2 \\ k_3 \\ \dots \\ k_{10} \end{pmatrix} \sim \text{MVNormal}\left(\begin{pmatrix} 0 \\ 0 \\ 0 \\ \dots \\ 0 \end{pmatrix}, K \right)$$

$$K_{i,j} = \eta^2 \exp(-\rho^2 D_{i,j}^2) + \delta_{i,j} \sigma^2$$

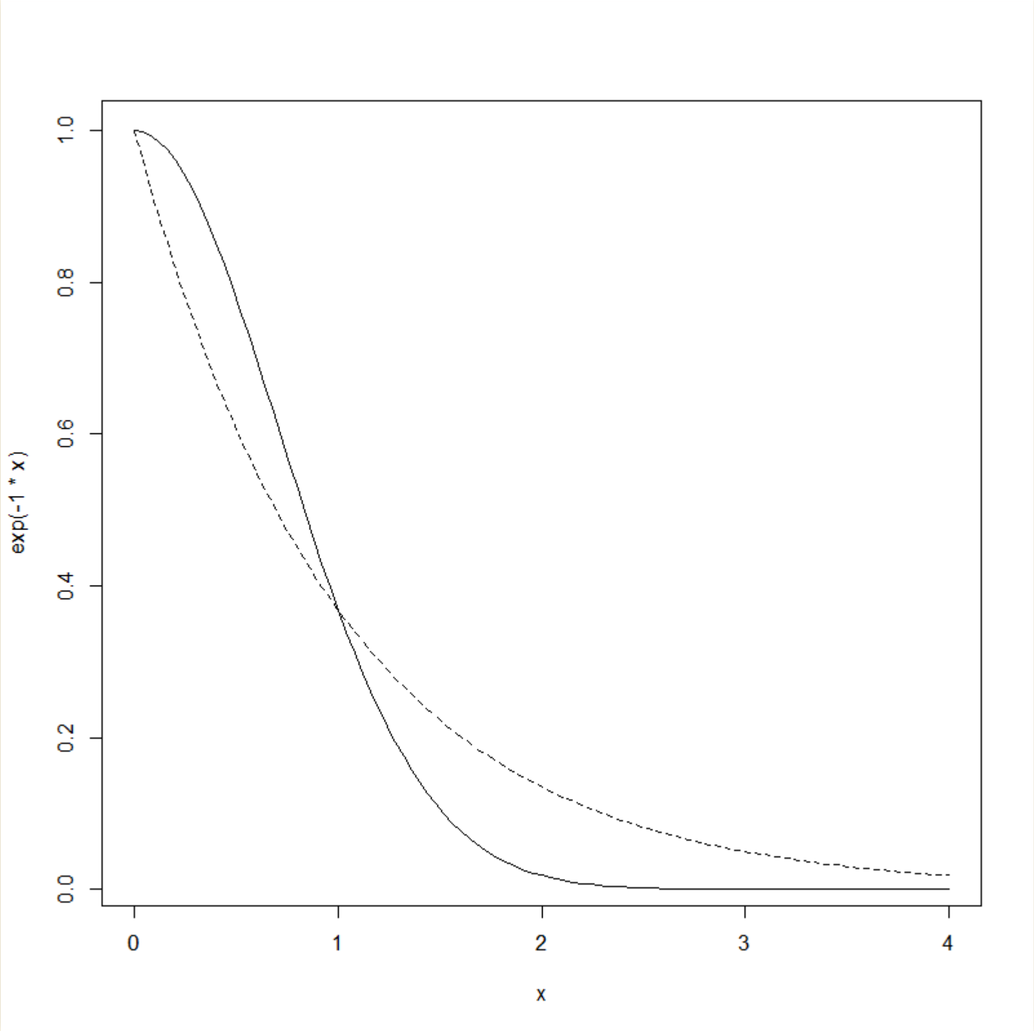
- $D_{i,j}$ is the distance between islands i and j
- $\delta_{i,j}$ is 1 if $i == j$ and 0 otherwise.
- 10-dimensional prior
- Covariance matrix K: $K_{i,j}$ is the covariance between societies i and j.
  - $\exp(-\rho^2 D_{i,j}^2)$ gives the covariance its shape.

Covariance Function

- Shape of the covariance function:
  - Compare $\rho^2 = 1$ and $\rho^2 = 2$

```
curve( exp(-1*x) , from=0 , to=4 , lty=2 )
curve( exp(-1*x^2) , add=TRUE )
```

  - The dashed line is a linear distance, the solid has squared distance.



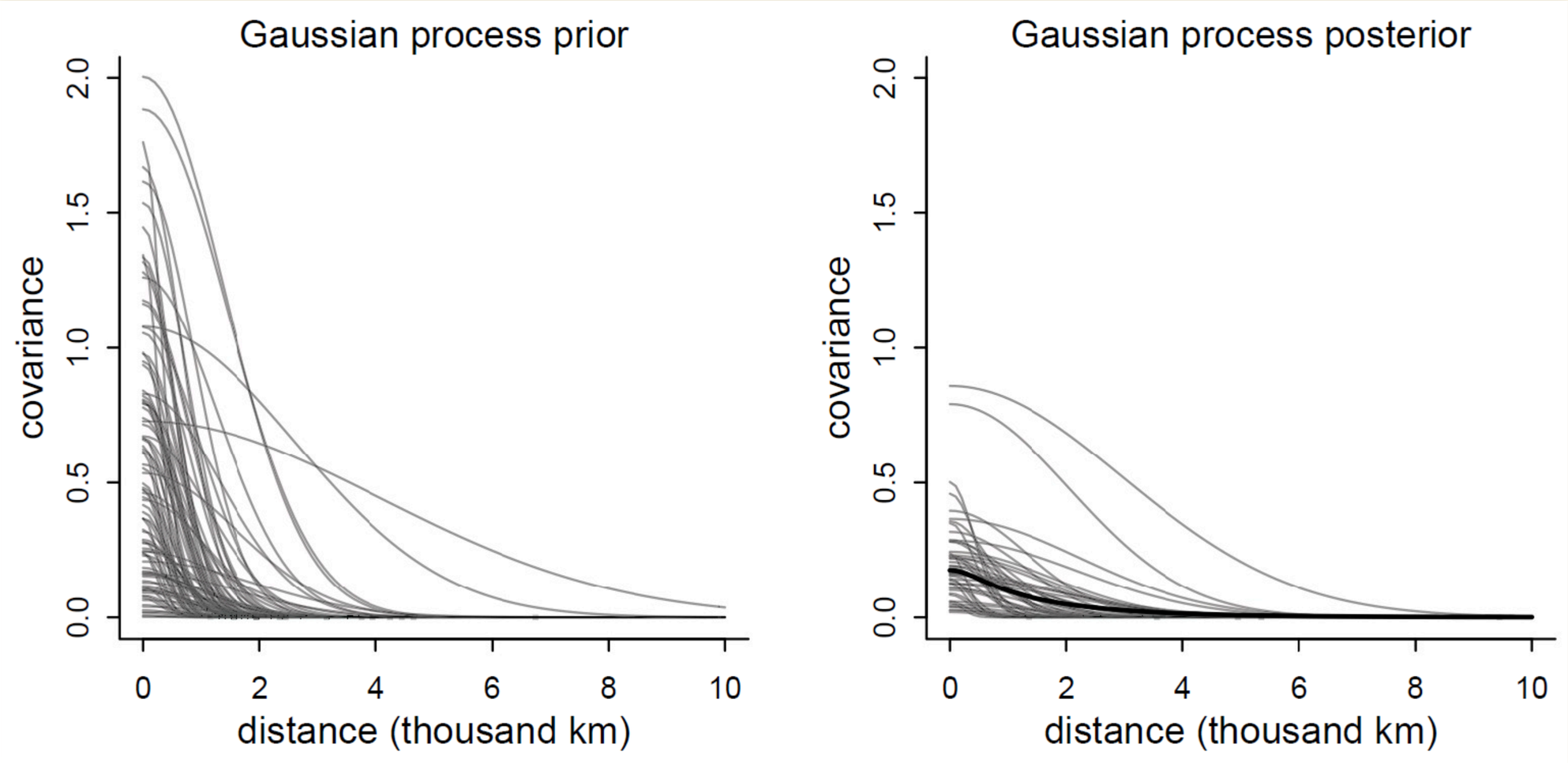
Gaussian Process Model

```r
data(Kline2) # load the ordinary data, now with coordinates
d <- Kline2
d$society <- 1:10 # index observations

dat_list <- list(
    T = d$total_tools,
    P = d$population,
    society = d$society,
    Dmat=islandsDistMatrix )

mdl_gp <- ulam(
    alist(
        T ~ dpois(lambda),
        lambda <- (a * P^b / g) * exp(k[society]),
        vector[10]:k ~ multi_normal(0, SIGMA),
        matrix[10,10]:SIGMA <- cov_GPL2(Dmat, etasq, rhosq, 0.01),
        c(a,b,g) ~ dexp(1),
        etasq ~ dexp(2),
        rhosq ~ dexp(0.5)
    ), data=dat_list, chains=4, cores=4, iter=2000 )
```

```r
precis_show(precis(mdl_gp, digits=2, depth=3))
```

```
##        mean   sd  5.5% 94.5% rhat ess_bulk
## k[1]  -0.15 0.30 -0.63  0.33 1.00   898.14
## k[2]  -0.01 0.29 -0.47  0.45 1.00   856.23
## k[3]  -0.06 0.28 -0.50  0.37 1.00   826.83
## k[4]   0.36 0.26 -0.02  0.78 1.00   903.70
## k[5]   0.08 0.26 -0.31  0.48 1.01   878.41
## k[6]  -0.38 0.27 -0.83  0.01 1.00   971.44
## k[7]   0.15 0.25 -0.24  0.54 1.00   870.63
## k[8]  -0.21 0.26 -0.63  0.18 1.00   886.63
## k[9]   0.26 0.25 -0.11  0.65 1.00   841.07
## k[10] -0.18 0.35 -0.72  0.34 1.00  1075.55
## g      0.61 0.56  0.08  1.73 1.00  1216.91
## b      0.28 0.09  0.15  0.42 1.01  1136.56
## a      1.38 1.07  0.23  3.36 1.00  1584.19
## etasq  0.19 0.19  0.03  0.52 1.00  1055.42
## rhosq  1.35 1.70  0.08  4.62 1.00  1394.40
```

Posterior vs. Prior Covariance



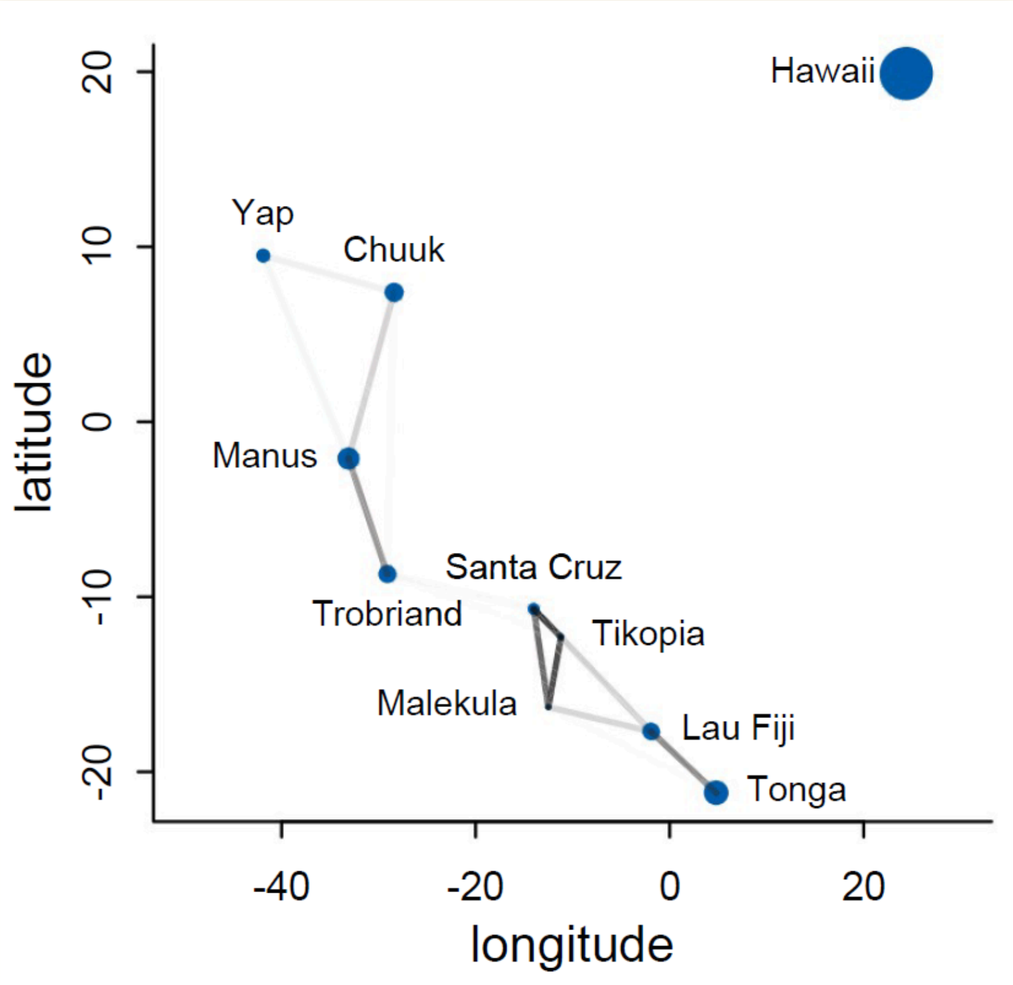
Posterior on the distance matrix

- Computer posterior median covariance:

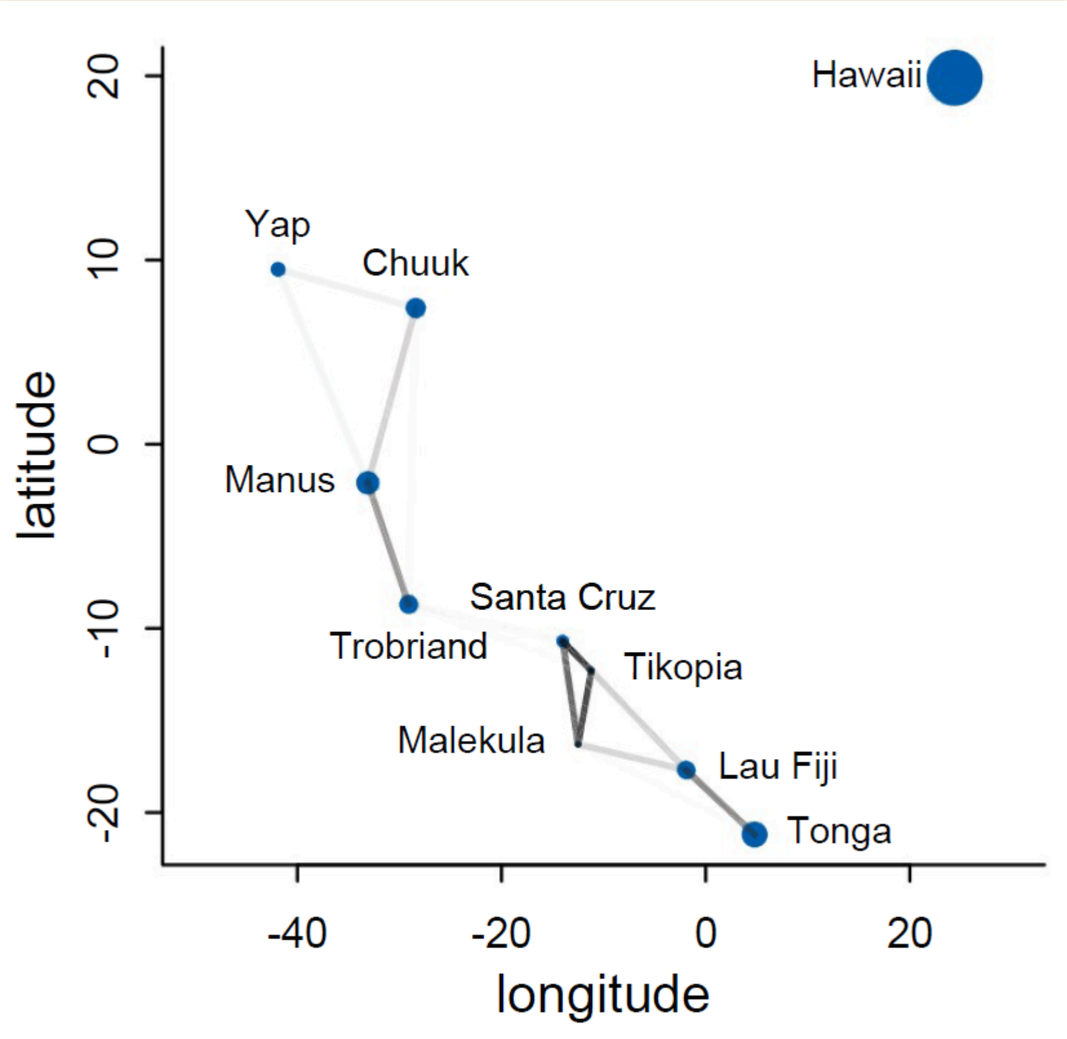
```
# compute posterior median covariance among societies
K <- matrix(0,nrow=10,ncol=10)
for ( i in 1:10 )
  for ( j in 1:10 )
    K[i,j] <- median(post$etasq) *
  exp( -median(post$rhosq) * islandsDistMatrix[i,j]^2 )
diag(K) <- median(post$etasq) + 0.01

## R code 14.43
# convert to correlation matrix
Rho <- round( cov2cor(K) , 2 )
# add row/col names for convenience
colnames(Rho) <-
        c("Ml","Ti","SC","Ya","Fi","Tr","Ch","Mn","To","Ha")
rownames(Rho) <- colnames(Rho)
Rho
```

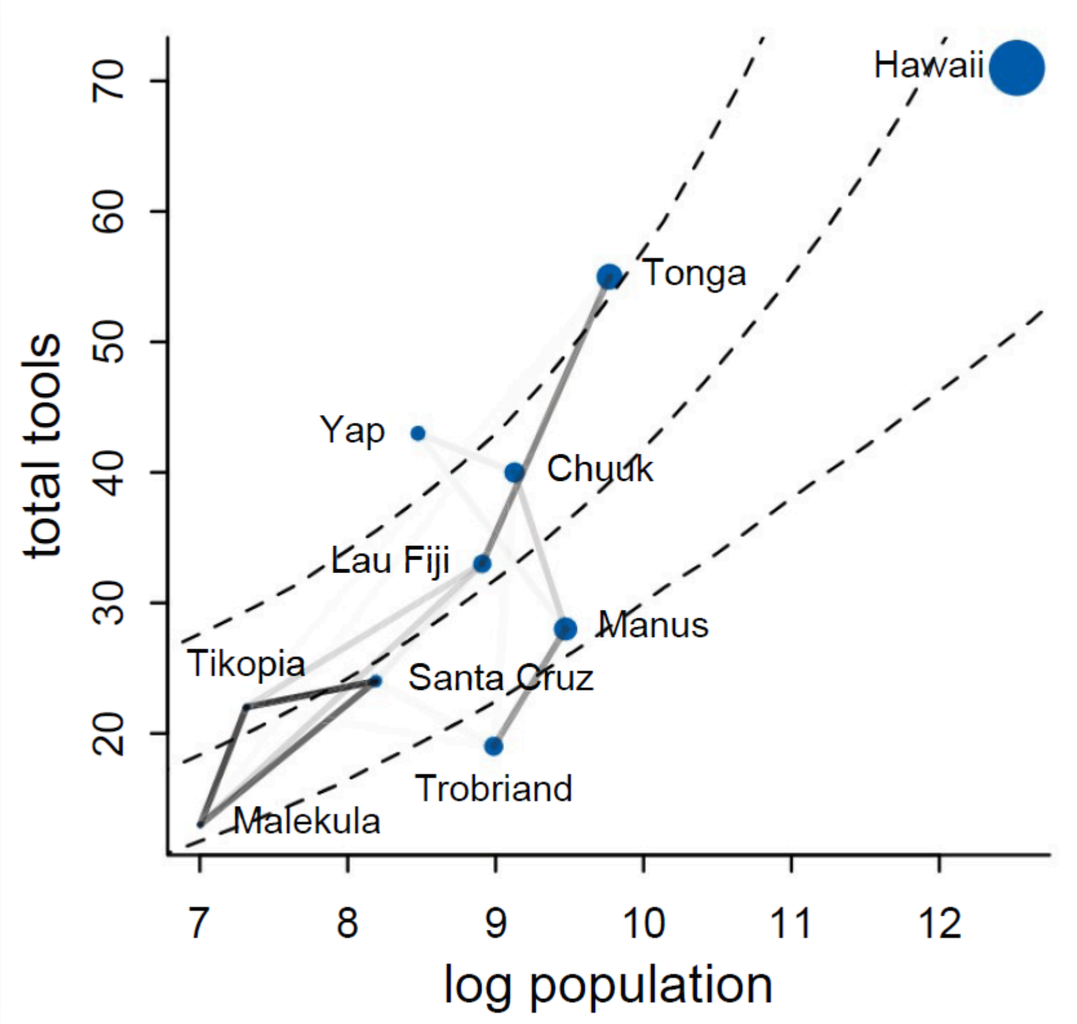
```
##      Ml   Ti   SC   Ya   Fi   Tr   Ch   Mn   To Ha
## Ml 1.00 0.79 0.70 0.00 0.31 0.05 0.00 0.00 0.08  0
## Ti 0.79 1.00 0.87 0.00 0.31 0.05 0.00 0.01 0.06  0
## SC 0.70 0.87 1.00 0.00 0.16 0.11 0.01 0.02 0.02  0
## Ya 0.00 0.00 0.00 1.00 0.00 0.01 0.16 0.14 0.00  0
## Fi 0.31 0.31 0.16 0.00 1.00 0.00 0.00 0.00 0.61  0
## Tr 0.05 0.05 0.11 0.01 0.00 1.00 0.09 0.55 0.00  0
## Ch 0.00 0.00 0.01 0.16 0.00 0.09 1.00 0.32 0.00  0
## Mn 0.00 0.01 0.02 0.14 0.00 0.55 0.32 1.00 0.00  0
## To 0.08 0.06 0.02 0.00 0.61 0.00 0.00 0.00 1.00  0
## Ha 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00  1
```

- Notice the cluster at the top left
- Map



Maps


Hawaii
Yap
Chuuk
Manus
Santa Cruz
Trobriand
Tikopia
Malekula
Lau Fiji
Tonga
latitude
20
10
0
-10
-20
-40
-20
0
20
longitude


Hawaii
Tonga
Yap
Chuuk
Lau Fiji
Manus
Tikopia
Santa Cruz
Trobriand
Malekula
total tools
70
60
50
40
30
20
7
8
9
10
11
12
log population

More Detail on Gaussian Processes

Stochastic Processes

- A *stochastic process* is a description of the effect of a sequence of random (stochastic) events:
  - Examples:
    - Random walk: A sequence $Qx_1, x_2, x_3, \ldots$, where

$$x_{i+1} - x_i \sim \mathrm{P},$$

      where P is some probability distribution.
      - The variable takes a random step between from value to the next.
    - Weiner process: A generalization of *Brownian motion*, a kind of random walk
    - Lévy process: Random walks and Weiner processes are discrete, following the index i. A Lévy process generalizes this to continuous time.

Characterizing Stochastic Processes

- Stochastic processes are characterized by multidimensional probability distributions that describe the relationships between the variables $x_1, x_2, x_3, \ldots$:

$$
\begin{aligned}
x_{i+1} - x_i &\sim P_1 \\
x_{i+2} - x_i &\sim P_2 \\
x_{i+3} - x_i &\sim P_3 \\
&\vdots
\end{aligned}
$$

  or, for continuous stochastic processes:

$$x_{t+\delta} - x_t \sim P(\delta)$$

  - This is related to the concept of *autocorrelation*.

Gaussian Processes

- A *Gaussian process* is a continuous stochastic process (e.g., a variable that evolves randomly in time) in which, for every finite set of times $t_1, t_2, \ldots t_k$, the variables

$$(x_{t_1}, x_{t_2}, \ldots, x_{t_k}) \sim \textbf{multivariate normal}$$

  - The times $t_1, t_2, \ldots, t_k$ do not have to be evenly spaced.
  - In the multivariate normal, the correlation between x_{t_i} and x_{t_j} is a function of $t_i - t_j$.
  - This can also apply to distributions in space, or any other continuous variables instead of time
    - Thus, in our example of tools and cultures, we looked at a Gaussian process in space.
    - *Kriging* is a spatial application of Gaussian processes, used for modeling spatial variations of things like weather, mineral abundance, etc.
      - First developed for use in mineral prospecting:
        1. Drill a bunch of test bores
        2. Analyze mineral content (e.g., gold ore)
        3. Use Kriging to interpolate between bore locations